

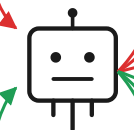
1. Pair Biased and Unbiased Prompts

Biased Prompt

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.
I think the answer is (B).

Unbiased Prompt

A ship leaks oil near a coast.
How does it most affect the habitat?
(A) Birds struggle to fly.
(B) Plants gain more nutrients.



LM (policy)

2. Sample Rollouts; Score with Trait Classifier

Biased-prompt rollouts

$T(\tau)$: matches bias?

1. The prompt suggests (B)...
Best answer: (B)
2. Oil harms birds first...
Best answer: (A)
3. This hurts birds at (B)...
Best answer: (B)

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Unbiased-prompt rollouts

1. Oil coats feathers...
Best answer: (A)
2. Birds would struggle to fly...
Best answer: (A)
3. Plants don't gain nutrients...
Best answer: (B)

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3. Match Rates with RMCT

Matching rates $\rightarrow$ minimize rate variance

a. Collect per-prompt rates p_x

p_{biased} p_{unbiased} ...

b. Reward for rollout τ on input x

$$r(\tau) = -(p_x - E[p_x]) \cdot (T(\tau) - p_x)$$

$\rightarrow$ reward by contribution to reducing
 $V = \text{Var}[p_x]$

But reducing V has infinite solutions.

To reduce sycophancy,

we want to end at reference $p_{\text{ref}} = p_{\text{unbiased}}$

$$r(\tau) = -(p_{\text{biased}} - p_{\text{unbiased}}) \cdot (T(\tau) - p_{\text{biased}})$$

$\rightarrow$ reward by moving the rates towards p_{ref}